

Supplementary materials to: Codimension-two bifurcation analysis of a discrete conformable fractional-order prey-predator model with nonlinear harvesting

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Supplementary Details for Section 3.1

(1) The coefficients B_{00} , B_{01} , $l_i(\theta)$ ($i = 1, 2$), k_{ij} , r_{ij} ($2 \leq i + j \leq 3$) in (3.2):

$$\begin{aligned}
 B_{00} &= x_1^* \exp\left[\frac{h^q}{q} \left(\frac{\bar{\omega}}{\beta} - \frac{(\bar{\omega} + \omega_*)y_1^*}{k_* + \beta y_1^*}\right)\right] - x_1^*, \quad B_{01} = y_1^* \exp\left[\frac{h^q \gamma k_*}{q(k_* + \beta y_1^*)}\right] - y_1^*, \\
 l_1(\theta) &= \exp\left[\frac{h^q}{q} \left(\frac{\bar{\omega}}{\beta} - \frac{(\bar{\omega} + \omega_*)y_1^*}{k_* + \beta y_1^*}\right)\right], \quad l_2(\theta) = \exp\left[\frac{h^q \gamma k_*}{q(k_* + \beta y_1^*)}\right], \quad k_{10} = -\frac{h^q \omega x_1^*}{q(k + x_1^*)}, \\
 k_{01} &= 1 + \frac{x_1^* h^q}{q} \left(-1 + \frac{\omega y_1^*}{(k + x_1^*)^2} + \frac{\sigma}{(m + x_1^*)^2}\right), \quad r_{01} = \frac{h^q \gamma \beta (y_1^*)^2}{q(k + x_1^*)^2}, \quad r_{10} = 1 - \frac{h^q \gamma \beta y_1^*}{q(k + x_1^*)}, \\
 k_{02} &= \frac{l_1(\theta) h^2 q \omega^2 x_1^*}{2q^2(k + x_1^*)^2}, \quad k_{11} = \frac{l_1(\theta) x_1^* h^q \omega}{q(k + x_1^*)} \left[-\frac{h^q}{q} \left(-1 + \frac{\omega y_1^*}{(k + x_1^*)^2} + \frac{\sigma}{(m + x_1^*)^2}\right) - 1 + \frac{1}{k + x_1^*}\right] \\
 k_{20} &= \frac{h^q l_1(\theta)}{q} \left[\left(-1 + \frac{\omega y_1^*}{(k + x_1^*)^2} + \frac{\sigma}{(m + x_1^*)^2}\right) \left(1 + \frac{h^q x_1^*}{2q} \left(-1 + \frac{\omega y_1^*}{(k + x_1^*)^2} + \frac{\sigma}{(m + x_1^*)^2}\right)\right) \right. \\
 &\quad \left. + x_1^* \left(\frac{\omega y_1^*}{(k + x_1^*)^3} + \frac{\sigma}{(m + x_1^*)^3}\right)\right], \\
 r_{20} &= \frac{l_2(\theta) h^q \gamma \beta (y_1^*)^2}{2q(k + x_1^*)^3} \left[\frac{h^q \gamma \beta y_1^*}{q(k + x_1^*)} - 2\right], \quad r_{11} = \frac{l_2(\theta) h^q \gamma \beta y_1^*}{(k + x_1^*)^2} \left[2 - \frac{h^q \gamma \beta y_1^*}{q(k + x_1^*)^2}\right], \\
 r_{02} &= \frac{l_2(\theta) h^q \gamma \beta y_1^*}{2q(k + x_1^*)} \left[\frac{h^q \gamma \beta y_1^*}{q(k + x_1^*)} - 2\right], \quad k_{03} = \frac{l_1(\theta) (h^q)^2 x_1^* \omega^3}{6q^2(k + x_1^*)^3}, \\
 k_{30} &= \frac{l_1(\theta) (h^q)^2}{6q^2} \left(-1 + \frac{\omega y_1^*}{(k + x_1^*)^2} + \frac{\sigma}{(m + x_1^*)^2}\right)^2 \left(3 + \frac{h^q x_1^*}{q} \left(-1 + \frac{\omega y_1^*}{(k + x_1^*)^2} + \frac{\sigma}{(m + x_1^*)^2}\right)\right) \\
 &\quad - \frac{l_1(\theta) h^q}{q} \left(\frac{\omega y_1^*}{(k + x_1^*)^3} + \frac{\sigma}{(m + x_1^*)^3}\right) \left(1 + \frac{h^q x_1^*}{q} \left(-1 + \frac{\omega y_1^*}{(k + x_1^*)^2} + \frac{\sigma}{(m + x_1^*)^2}\right)\right) \\
 &\quad + \frac{l_1(\theta) h^q x_1^*}{q} \left(\frac{\omega y_1^*}{(k + x_1^*)^4} + \frac{\sigma}{(m + x_1^*)^4}\right),
 \end{aligned}$$

$$\begin{aligned}
k_{21} &= \frac{l_1(\theta)h^q\gamma\beta(y_1^*)^2}{2q(k+x_1^*)^2} \left[\left(\frac{h^q\gamma\beta}{q} + \frac{3}{(k+x_1^*)^2} \right) \left(\frac{h^q\gamma\beta y_1^*}{q(k+x_1^*)} - 2 \right) + \frac{h^q\gamma\beta y_1^*}{q(k+x_1^*)^3} \right], \\
k_{12} &= \frac{l_1(\theta)(h^q)^2\omega^2}{2q^2(k+x_1^*)^2} \left[\frac{(h^q)^2}{q^2} \left(-1 + \frac{\omega y_1^*}{(k+x_1^*)^2} + \frac{\sigma}{(m+x_1^*)^2} \right) + \frac{k-x_1^*}{k+x_1^*} \right], \\
r_{30} &= -\frac{l_2(\theta)h^q\gamma\beta(y_1^*)^2}{6q^2(k+x_1^*)^4} \left[\left(\frac{h^q\gamma\beta y_1^*}{q(k+x_1^*)} - 2 \right) \left(\frac{h^q\gamma\beta}{q(k+x_1^*)} - 3 \right) - \frac{h^q\gamma\beta y_1^*}{q(k+x_1^*)} \right], \\
r_{21} &= \frac{l_2(\theta)h^q\gamma\beta y_1^*}{2q(k+x_1^*)^3} \left[\left(\frac{h^q\gamma\beta y_1^*}{q(k+x_1^*)} - 2 \right) \left(2 - \frac{h^q\gamma\beta y_1^*}{q(k+x_1^*)} \right) + \frac{h^q\gamma\beta y_1^*}{q(k+x_1^*)} \right], \\
r_{12} &= -\frac{l_2(\theta)h^q\gamma\beta}{2q(k+x_1^*)^2} \left[\frac{h^q\gamma\beta y_1^*}{q(k+x_1^*)} \left(5 + \frac{h^q\gamma\beta y_1^*}{q(k+x_1^*)} \right) - 2 \right], \\
r_{03} &= \frac{l_2(\theta)(h^q)^2\gamma^2\beta^2}{6q^2(k+x_1^*)^2} \left[1 - \frac{h^q\gamma\beta}{q(k+x_1^*)} \left(\frac{h^q\gamma\beta y_1^*}{q(k+x_1^*)} - 2 \right) \right].
\end{aligned}$$

(2) The coefficients $\widehat{k}_{ij}(\theta)$ ($i+j=2$), $\widehat{r}_{ij}(\theta)$ ($0 \leq i+j \leq 2$) in (3.3):

$$\begin{aligned}
\widehat{k}_{00}(\theta) &= B_{01}, \quad \widehat{k}_{10}(\theta) = r_{10}(\theta)\beta + r_{01}(\theta) - 1, \quad \widehat{k}_{01}(\theta) = -\frac{\beta q}{h^q\gamma}r_{10} - 1, \quad \widehat{k}_{02}(\theta) = \frac{\beta^2 q^2}{(h^q)^2\gamma^2}r_{20}(\theta), \\
\widehat{k}_{20}(\theta) &= \beta^2 r_{20}(\theta) + \beta r_{11}(\theta) + r_{02}(\theta), \quad \widehat{k}_{11}(\theta) = -\frac{2\beta^2 q}{h^q\gamma}r_{20}(\theta) - \frac{\beta q}{h^q\gamma}r_{11}(\theta), \quad \widehat{r}_{00}(\theta) = B_{00} - \beta B_{01}, \\
\widehat{r}_{01}(\theta) &= -\frac{\beta q}{h^q\gamma}k_{10}(\theta) - \beta \left[-\frac{\beta q}{h^q\gamma}r_{10}(\theta) \right] - 1, \quad \widehat{r}_{10}(\theta) = \left[\beta k_{10}(\theta) + k_{01}(\theta) \right] - \beta \left[\beta r_{10}(\theta) + r_{01}(\theta) \right], \\
\widehat{r}_{20}(\theta) &= \left[\beta^2 k_{20}(\theta) + \beta k_{11}(\theta) + k_{02}(\theta) \right] - \beta \left[\beta^2 r_{20}(\theta) + \beta r_{11}(\theta) + r_{02}(\theta) \right], \\
\widehat{r}_{11}(\theta) &= \frac{\beta^2 q^2}{(h^q)^2\gamma^2}k_{20}(\theta) - \beta \frac{\beta^2 q^2}{(h^q)^2\gamma^2}r_{20}(\theta), \\
\widehat{r}_{02}(\theta) &= \left[-\frac{2\beta^2 q}{h^q\gamma}k_{20}(\theta) - \frac{\beta q}{h^q\gamma}r_{11}(\theta) \right] - \beta \left[-\frac{2\beta^2 q}{h^q\gamma}r_{20}(\theta) - \frac{\beta q}{h^q\gamma}r_{11}(\theta) \right].
\end{aligned}$$

(3) The coefficients $a_{ij}(\theta)$, $b_{ij}(\theta)$ ($0 \leq i+j \leq 2$) in (3.4):

$$\begin{aligned}
a_{00}(\theta) &= \widehat{k}_{00}(\theta) - \left(\frac{1}{2}\widehat{k}_{10}(\theta) - \frac{1}{3}\widehat{r}_{01}(\theta) \right) \widehat{k}_{00}(\theta) - \left(\frac{1}{2} - \frac{1}{3}\widehat{k}_{10}(\theta) + \frac{1}{2}\widehat{r}_{01}(\theta) \right) + \frac{1}{4}\widehat{r}_{10}(\theta) - \frac{1}{3}\widehat{r}_{01}(\theta) \widehat{k}_{00}(\theta), \\
a_{10}(\theta) &= \widehat{k}_{10}(\theta) - \frac{1}{2}\widehat{r}_{10}(\theta), \quad a_{01}(\theta) = \widehat{k}_{01}(\theta) - \frac{1}{2}\widehat{k}_{10}(\theta) + \frac{1}{3}\widehat{r}_{10}(\theta) - \frac{1}{2}\widehat{r}_{01}(\theta), \\
a_{20}(\theta) &= \widehat{k}_{20}(\theta) - \frac{1}{2}\widehat{r}_{20}(\theta), \quad b_{11}(\theta) = \widehat{k}_{11}(\theta) - \widehat{k}_{20}(\theta) + \frac{2}{3}\widehat{r}_{20}(\theta) - \frac{1}{2}\widehat{r}_{11}(\theta), \\
a_{02}(\theta) &= \widehat{k}_{02}(\theta) + \frac{1}{6}\widehat{k}_{20}(\theta) - \frac{1}{2}\widehat{k}_{11}(\theta) - \frac{1}{6}\widehat{r}_{20}(\theta) + \frac{1}{3}\widehat{r}_{11}(\theta) - \frac{1}{2}\widehat{r}_{02}(\theta), \\
b_{00}(\theta) &= \widehat{r}_{00}(\theta) - \frac{1}{2}\widehat{r}_{10}(\theta)\widehat{k}_{00}(\theta) + \left(\frac{1}{3}\widehat{r}_{10}(\theta) - \frac{1}{2}\widehat{r}_{01}(\theta) \right) \widehat{r}_{00}(\theta), \\
b_{10}(\theta) &= \widehat{r}_{10}(\theta), \quad a_{01}(\theta) = \widehat{r}_{01}(\theta) - \frac{1}{2}\widehat{r}_{10}(\theta), \quad b_{20}(\theta) = \widehat{r}_{20}(\theta), \\
b_{11}(\theta) &= \widehat{r}_{11}(\theta) - \frac{1}{2}\widehat{r}_{20}(\theta), \quad b_{02}(\theta) = \widehat{r}_{02}(\theta) + \frac{1}{6}\widehat{r}_{20}(\theta) - \frac{1}{2}\widehat{r}_{11}(\theta).
\end{aligned}$$

(4) The coefficients $\beta_i(\theta)$ ($i = 1, 2, 3, 4$) in (3.7):

$$\begin{aligned}\beta_1(\theta) &= \widehat{r}_{00}(\theta) + \frac{1}{2} \left(\frac{1}{3} \widehat{k}_{20}(0) - \widehat{k}_{11}(0) + 2\widehat{k}_{02}(0) - \frac{1}{4} \widehat{r}_{20}(0) + \frac{5}{12} \widehat{r}_{11}(0) \right. \\ &\quad \left. - \frac{1}{2} \widehat{r}_{02}(0) \right) \widehat{r}_{00}^2(\theta) + \frac{5}{12} \widehat{r}_{00}(\theta) \widehat{r}_{10}(\theta) - \frac{1}{2} \widehat{r}_{00}(\theta) \widehat{r}_{01}(\theta), \\ \beta_2(\theta) &= -\widehat{r}_{10}(\theta) + \widehat{r}_{01}(\theta) - \left(\frac{1}{6} \widehat{k}_{20}(0) + \frac{1}{2} \widehat{k}_{11}(0) - 2\widehat{k}_{02}(0) - \frac{1}{6} \widehat{r}_{20}(0) + \frac{1}{12} \widehat{r}_{11}(0) \right) \widehat{r}_{00}(\theta), \\ \beta_3(\theta) &= \widehat{r}_{10}(\theta) - \left(\widehat{k}_{20}(0) - \widehat{k}_{11}(0) - \frac{1}{6} \widehat{r}_{20}(0) \right) \widehat{r}_{00}(\theta), \\ \beta_4(\theta) &= \widehat{r}_{10}(\theta) + \left(\widehat{k}_{20}(0) - \widehat{k}_{11}(0) - \frac{3}{2} \widehat{r}_{20}(0) + 2\widehat{r}_{11}(0) - 2\widehat{r}_{02}(0) \right) \widehat{r}_{00}(\theta).\end{aligned}$$

Supplementary Details for Section 3.2

(1) The coefficients $D_{ij}(\theta)$ ($i, j = 1, 2$) in (3.9) are given by:

$$\begin{aligned}D_{11}(\theta) &= \left[1 + \frac{h^q x_1^*}{q} \left(-1 + \frac{\bar{\omega} y_1^*}{(\bar{k} + x_1^*)^2} + \frac{\sigma}{(m + x_1^*)^2} \right) \right] (l_1(\theta) - 1) \\ &\quad + \frac{h^q x_1^*}{q} \left[\frac{(\bar{\omega} + \omega_*) y_1^*}{(\bar{k} + k_* + x_1^*)^2} - \frac{\bar{\omega} y_1^*}{(\bar{k} + x_1^*)^2} \right] l_1(\theta), \\ D_{12}(\theta) &= -\frac{h^q \bar{\omega} x_1^*}{q(\bar{k} + x_1^*)} (l_1(\theta) - 1) + \left[\frac{h^q \bar{\omega} x_1^*}{q(\bar{k} + x_1^*)} - \frac{h^q (\bar{\omega} + \omega_*) x_1^*}{q(\bar{k} + k_* + x_1^*)} \right] l_1(\theta), \\ D_{21}(\theta) &= \frac{h^q \gamma \beta (y_1^*)^2}{q(\bar{k} + x_1^*)^2} (l_2(\theta) - 1) + \left[\frac{h^q \gamma \beta (y_1^*)^2}{q(\bar{k} + k_* + x_1^*)^2} - \frac{h^q \gamma \beta (y_1^*)^2}{q(\bar{k} + x_1^*)^2} \right] l_2(\theta), \\ D_{22}(\theta) &= \left[1 - \frac{h^q \gamma \beta y_1^*}{q(\bar{k} + x_1^*)} \right] (l_2(\theta) - 1) + \frac{h^q \gamma \beta y_1^*}{q} \left[\frac{1}{\bar{k} + x_1^*} - \frac{1}{\bar{k} + k_* + x_1^*} \right] l_2(\theta).\end{aligned}$$

Expanding $l_1(\theta)$ and $l_2(\theta)$ as Taylor series at the origin $(\omega_*, k_*) = (0, 0)$ yields:

$$l_1(\theta) = 1 - \frac{h^q}{q\beta} \omega_* + \frac{\bar{\omega}}{\beta^2 y_*} k_* + O(|\theta|^2), \quad l_2(\theta) = 1 - \frac{h^q \gamma}{q\beta y_*} k_* + O(|\theta|^2).$$

Obviously, $D_{11}(0) = D_{12}(0) = D_{21}(0) = D_{22}(0) = 0$.

(2) The coefficients $\widehat{k}_{ij}(\theta)$, $\widehat{r}_{ij}(\theta)$ ($2 \leq i + j \leq 3$), $\widehat{D}_{ij}(\theta)$ ($i, j = 1, 2$) in (3.10):

$$\begin{aligned}\widehat{D}_{11}(\theta) &= \frac{1}{2} [D_{11}(\alpha + 1) + D_{12}\psi] - \frac{\alpha - 1}{2\psi} D_{21}(\alpha + 1) + D_{22}\psi, \\ \widehat{D}_{12}(\theta) &= \frac{1}{2} [D_{11}(\alpha - 1) + D_{12}\psi] - \frac{\alpha - 1}{2\psi} D_{21}(\alpha - 1) + D_{22}\psi, \\ \widehat{D}_{21}(\theta) &= -\frac{1}{2} [D_{11}(\alpha + 1) + D_{12}\psi] + \frac{\alpha + 1}{2\psi} D_{21}(\alpha + 1) + D_{22}\psi, \\ \widehat{D}_{22}(\theta) &= -\frac{1}{2} [D_{11}(\alpha - 1) + D_{12}\psi] + \frac{\alpha + 1}{2\psi} D_{21}(\alpha - 1) + D_{22}\psi.\end{aligned}$$

$$\begin{aligned}
\tilde{k}_{20}(\theta) &= \left(\frac{1}{2}k_{20} - \frac{\alpha-1}{2\psi}r_{20} \right) (\alpha+1)^2 + \left(\frac{1}{2}k_{11} - \frac{\alpha-1}{2\psi}r_{11} \right) (\alpha+1)\psi \\
&\quad + \left(\frac{1}{2}k_{02} - \frac{\alpha-1}{2\psi}r_{02} \right) \psi^2, \\
\tilde{k}_{11}(\theta) &= 2 \left(\frac{1}{2}k_{20} - \frac{\alpha-1}{2\psi}r_{20} \right) (\alpha^2-1) + 2 \left(\frac{1}{2}k_{11} - \frac{\alpha-1}{2\psi}r_{11} \right) \alpha\psi \\
&\quad + 2 \left(\frac{1}{2}k_{02} - \frac{\alpha-1}{2\psi}r_{02} \right) \psi^2, \\
\tilde{k}_{02}(\theta) &= \left(\frac{1}{2}k_{20} - \frac{\alpha-1}{2\psi}r_{20} \right) (\alpha-1)^2 + \left(\frac{1}{2}k_{11} - \frac{\alpha-1}{2\psi}r_{11} \right) (\alpha-1)\psi \\
&\quad + \left(\frac{1}{2}k_{02} - \frac{\alpha-1}{2\psi}r_{02} \right) \psi^2, \\
\tilde{k}_{30}(\theta) &= \left(\frac{1}{2}k_{30} - \frac{\alpha-1}{2\psi}r_{30} \right) (1+\alpha)^3 + \left(\frac{1}{2}k_{03} - \frac{\alpha-1}{2\psi}r_{03} \right) \psi^3 \\
&\quad + \left(\frac{1}{2}k_{12} - \frac{\alpha-1}{2\psi}r_{12} \right) (1+\alpha)\psi^2 + \left(\frac{1}{2}k_{21} - \frac{\alpha-1}{2\psi}r_{21} \right) (1+\alpha)^2\psi, \\
\tilde{k}_{21}(\theta) &= 3 \left(\frac{1}{2}k_{30} - \frac{\alpha-1}{2\psi}r_{30} \right) (1+\alpha)^2(\alpha-1) + \left(\frac{1}{2}k_{03} - \frac{\alpha-1}{2\psi}r_{03} \right) \psi^3 \\
&\quad + \left(\frac{1}{2}k_{21} - \frac{\alpha-1}{2\psi}r_{21} \right) [2(\alpha^2-1)\psi + (1+\alpha)^2\psi] \\
&\quad + \left(\frac{1}{2}k_{12} - \frac{\alpha-1}{2\psi}r_{12} \right) [2(1+\alpha)\psi^2 + (\alpha-1)\psi^2], \\
\tilde{k}_{12}(\theta) &= 3 \left(\frac{1}{2}k_{30} - \frac{\alpha-1}{2\psi}r_{30} \right) (1+\alpha)(\alpha-1)^2 + \frac{3}{2}r_{03}\psi^3 \\
&\quad + \left(\frac{1}{2}k_{21} - \frac{\alpha-1}{2\psi}r_{21} \right) [2(\alpha^2-1)\psi + (\alpha-1)^2\psi] \\
&\quad + \left(\frac{1}{2}k_{12} - \frac{\alpha-1}{2\psi}r_{12} \right) [(1+\alpha)\psi^2 + 2(\alpha-1)\psi^2], \\
\tilde{k}_{03}(\theta) &= \left(\frac{1}{2}k_{30} - \frac{\alpha-1}{2\psi}r_{30} \right) (\alpha-1)^3 + \left(\frac{1}{2}k_{03} - \frac{\alpha-1}{2\psi}r_{03} \right) \psi^3 \\
&\quad + \left(\frac{1}{2}k_{21} - \frac{\alpha-1}{2\psi}r_{21} \right) (\alpha-1)^2\psi + \left(\frac{1}{2}k_{12} - \frac{\alpha-1}{2\psi}r_{12} \right) (\alpha-1)\psi^2.
\end{aligned}$$

and $\tilde{r}_{ij}(\theta)$ ($2 \leq i+j \leq 3$) are obtained by replacing $\frac{1}{2}k_{ij}$, $-\frac{\alpha-1}{2\psi}r_{ij}$ in the right-hand side of $\tilde{r}_{ij}(\theta)$ ($2 \leq i+j \leq 3$) with $-\frac{1}{2}k_{ij}$ and $\frac{\alpha+1}{2\psi}r_{ij}$, respectively.

(3) The coefficients $\bar{k}_{ij}(\theta)$, $\bar{r}_{ij}(\theta)$ ($2 \leq i+j \leq 3$), $\sigma_1(\theta)$, and $\sigma_2(\theta)$ in (3.11):

$$\begin{aligned}
\sigma_1(\theta) &= \frac{1 + \xi_2(\theta)}{2(1 - \xi_1(\theta)\xi_2(\theta))} B_{00}(\theta) - \frac{(\alpha-1) + \xi_2(\theta)(\alpha+1)}{2\psi(1 - \xi_1(\theta)\xi_2(\theta))} B_{01}(\theta), \\
\sigma_2(\theta) &= -\frac{1 + \xi_1(\theta)}{2(1 - \xi_1(\theta)\xi_2(\theta))} B_{00}(\theta) + \frac{(\alpha+1) + \xi_1(\theta)(\alpha-1)}{2\psi(1 - \xi_1(\theta)\xi_2(\theta))} B_{01}(\theta).
\end{aligned}$$

$$\begin{aligned}
\bar{k}_{20}(\theta) &= \frac{2}{1 - \xi_1(\theta)\xi_2(\theta)} \left[(\tilde{k}_{20} - \xi_2\tilde{r}_{20}) + (\tilde{k}_{11} - \xi_2\tilde{r}_{11})\xi_1 + (\tilde{k}_{02} - \xi_2\tilde{r}_{02})\xi_1^2 \right], \\
\bar{k}_{11}(\theta) &= \frac{1}{1 - \xi_1(\theta)\xi_2(\theta)} \left[2(\tilde{k}_{20} - \xi_2\tilde{r}_{20})\xi_2 + (\tilde{k}_{11} - \xi_2\tilde{r}_{11})(1 + \xi_1\xi_2) + 2(\tilde{k}_{02} - \xi_2\tilde{r}_{02})\xi_1 \right], \\
\bar{k}_{02}(\theta) &= \frac{2}{1 - \xi_1(\theta)\xi_2(\theta)} \left[(\tilde{k}_{20} - \xi_2\tilde{r}_{20})\xi_2^2 + (\tilde{k}_{11} - \xi_2\tilde{r}_{11})\xi_2 + (\tilde{k}_{02} - \xi_2\tilde{r}_{02}) \right], \\
\bar{k}_{30}(\theta) &= \frac{6}{1 - \xi_1(\theta)\xi_2(\theta)} \left[(\tilde{k}_{30} - \xi_2\tilde{r}_{30}) + (\tilde{k}_{21} - \xi_2\tilde{r}_{21})\xi_1 + (\tilde{k}_{12} - \xi_2\tilde{r}_{12})\xi_1^2 + (\tilde{k}_{03} - \xi_2\tilde{r}_{03})\xi_1^3 \right], \\
\bar{k}_{21}(\theta) &= \frac{2}{1 - \xi_1(\theta)\xi_2(\theta)} \left[3(\tilde{k}_{30} - \xi_2\tilde{r}_{30})\xi_2 + (\tilde{k}_{21} - \xi_2\tilde{r}_{21})(1 + 2\xi_1\xi_2) \right. \\
&\quad \left. + (\tilde{k}_{12} - \xi_2\tilde{r}_{12})(2\xi_1^2\xi_2 + 2\xi_1) + 3(\tilde{k}_{03} - \xi_2\tilde{r}_{03})\xi_1 \right], \\
\bar{k}_{12}(\theta) &= \frac{2}{1 - \xi_1(\theta)\xi_2(\theta)} \left[3(\tilde{k}_{30} - \xi_2\tilde{r}_{30})\xi_2^2 + (\tilde{k}_{21} - \xi_2\tilde{r}_{21})(\xi_1\xi_2^2 + 2\xi_2) \right. \\
&\quad \left. + (\tilde{k}_{12} - \xi_2\tilde{r}_{12})(1 + 2\xi_1\xi_2) + 3(\tilde{k}_{03} - \xi_2\tilde{r}_{03})\xi_1 \right], \\
\bar{k}_{03}(\theta) &= \frac{6}{1 - \xi_1(\theta)\xi_2(\theta)} \left[(\tilde{k}_{30} - \xi_2\tilde{r}_{30})\xi_2^3 + (\tilde{k}_{21} - \xi_2\tilde{r}_{21})\xi_2^2 + (\tilde{k}_{12} - \xi_2\tilde{r}_{12})\xi_2 + (\tilde{k}_{03} - \xi_2\tilde{r}_{03}) \right].
\end{aligned}$$

and $\bar{r}_{ij}(\theta)$ ($2 \leq i+j \leq 3$) are obtained by replacing $(\tilde{k}_{ij} - \xi_2\tilde{r}_{ij})$ in $\bar{k}_{ij}(\theta)$ ($2 \leq i+j \leq 3$) with $(-\xi_1\tilde{k}_{ij} + \tilde{r}_{ij})$.

(4) The coefficients in (3.12) are given by:

$$\begin{aligned}
A_1 &= -\frac{h^q x_1^*}{2q\beta}, \quad A_2 = \frac{h^q x_1^*}{2q\beta^2(y_1^*)^2} - \frac{h^q \gamma(\alpha - 1)}{2\psi\beta}, \quad B_1 = \frac{h^q x_1^*}{2q\beta}, \quad B_2 = -\frac{h^q x_1^*}{2q\beta(y_1^*)^2} + \frac{h^q \gamma(\alpha + 1)}{2\psi\beta}, \\
A_3 &= \frac{1}{2} \left[\frac{(\alpha + 1)h^q}{q} \left(\frac{x_1^*}{y_1^*\beta} - \frac{1}{\beta} \left(1 + \frac{h^q x_1^*}{q} \left(-1 + \frac{\bar{\omega}}{\beta y_1^*} + \frac{\sigma}{(m + x_1^*)^2} \right) \right) \right) + \frac{\psi h^q x_1^*}{q\beta y_1^*} \left(1 + \frac{h^q \bar{\omega}}{q\beta} \right) \right], \\
A_4 &= \frac{1}{2} \left[\frac{(\alpha + 1)\bar{\omega}}{\beta^2 y_1^*} \left(1 + \frac{h^q x_1^*}{q} \left(-1 + \frac{\bar{\omega}}{\beta y_1^*} + \frac{\sigma}{(m + x_1^*)^2} \right) - \frac{2h^q x_1^*}{q\beta} \right) + \frac{\psi h^q x_1^* \bar{\omega}}{q\beta y_1^*} \left(\frac{1}{\beta y_1^*} - \frac{\bar{\omega}}{y_1^* \beta^2} \right) \right] \\
&\quad - \frac{\alpha - 1}{2\psi} \left[\frac{(\alpha - 1)h^q \gamma}{q\beta} \left(\frac{h^q \gamma}{q\beta} - \frac{2}{\beta y_1^*} \right) + \frac{\psi h^q \gamma}{q} \left(\frac{1}{\beta y_1^*} \left(1 - \frac{h^q \gamma}{q} \right) + \frac{1}{\beta y_1^*} \right) \right], \\
B_3 &= -\frac{1}{2} \left[\frac{(\alpha - 1)h^q}{q} \left(\frac{x_1^*}{y_1^*\beta} - \frac{1}{\beta} \left(1 + \frac{h^q x_1^*}{q} \left(-1 + \frac{\bar{\omega}}{\beta y_1^*} + \frac{\sigma}{(m + x_1^*)^2} \right) \right) \right) + \frac{\psi h^q x_1^*}{2q\beta y_1^*} \left(1 + \frac{h^q \bar{\omega}}{q\beta} \right) \right], \\
B_4 &= -\frac{1}{2} \left[\frac{(\alpha - 1)\bar{\omega}}{\beta^2 y_1^*} \left(1 + \frac{h^q x_1^*}{q} \left(-1 + \frac{\bar{\omega}}{\beta y_1^*} + \frac{\sigma}{(m + x_1^*)^2} \right) - \frac{2h^q x_1^*}{q\beta} \right) + \frac{\psi h^q x_1^* \bar{\omega}}{q\beta y_1^*} \left(\frac{1}{\beta y_1^*} - \frac{\bar{\omega}}{y_1^* \beta^2} \right) \right] \\
&\quad + \frac{\alpha + 1}{2\psi} \left[\frac{(\alpha - 1)h^q \gamma}{q\beta} \left(\frac{h^q \gamma}{q\beta} - \frac{2}{\beta y_1^*} \right) + \frac{\psi h^q \gamma}{q} \left(\frac{1}{\beta y_1^*} \left(1 - \frac{h^q \gamma}{q} \right) + \frac{1}{\beta y_1^*} \right) \right].
\end{aligned}$$

Supplementary Details for Section 3.3

(1) The coefficients $\hat{\sigma}_{ij}(\theta), \hat{\zeta}_{ij}(\theta)$ ($2 \leq i+j \leq 3$), and $\tilde{D}_{ij}(\theta)$ ($i, j = 1, 2$) in (3.17):

$$\begin{aligned}
\tilde{D}_{11}(\theta) &= \hat{\xi}_1 D_{21} + D_{22}, \quad \tilde{D}_{12}(\theta) = \hat{\xi}_2 D_{21}, \\
\tilde{D}_{21}(\theta) &= \hat{\xi}_3(\hat{\xi}_1 D_{11} + D_{12}) + \hat{\xi}_4(\hat{\xi}_1 D_{21} + D_{22}), \quad \tilde{D}_{22}(\theta) = \hat{\xi}_2(\hat{\xi}_3 D_{11} + \hat{\xi}_4 D_{21}).
\end{aligned}$$

$$\begin{aligned}
\hat{\sigma}_{20}(\theta) &= r_{20}\hat{\xi}_1^2 + r_{11}\hat{\xi}_1 + r_{02}, \hat{\sigma}_{11}(\theta) = 2r_{20}\hat{\xi}_1\hat{\xi}_2 + r_{11}\hat{\xi}_2, \\
\hat{\sigma}_{02}(\theta) &= r_{20}\hat{\xi}_2, \hat{\sigma}_{30}(\theta) = r_{30}\hat{\xi}_1^3 + r_{21}\hat{\xi}_1^2 + r_{12}\hat{\xi}_1 + r_{03}, \\
\hat{\sigma}_{21}(\theta) &= 3r_{30}\hat{\xi}_1^2\hat{\xi}_2 + 2r_{21}\hat{\xi}_1\hat{\xi}_2 + r_{12}\hat{\xi}_2, \hat{\sigma}_{12}(\theta) = 3r_{30}\hat{\xi}_1\hat{\xi}_2^2 + r_{21}\hat{\xi}_2^2, \\
\hat{\sigma}_{03}(\theta) &= r_{30}\hat{\xi}_2^3, \hat{\zeta}_{20}(\theta) = \hat{\xi}_3(k_{20}\hat{\xi}_1^2 + k_{11}\hat{\xi}_1 + k_{02}) + \hat{\xi}_4(r_{20}\hat{\xi}_1^2 + r_{11}\hat{\xi}_1 + r_{02}), \\
\hat{\zeta}_{11}(\theta) &= \hat{\xi}_3(2k_{20}\hat{\xi}_1\hat{\xi}_2 + k_{11}\hat{\xi}_2) + \hat{\xi}_4(2r_{20}\hat{\xi}_1\hat{\xi}_2 + r_{11}\hat{\xi}_2), \hat{\zeta}_{02}(\theta) = \hat{\xi}_2(k_{20}\hat{\xi}_3 + r_{20}\hat{\xi}_4), \\
\hat{\zeta}_{30}(\theta) &= \hat{\xi}_3(k_{30}\hat{\xi}_1^3 + k_{21}\hat{\xi}_1^2 + k_{12} + k_{03}) + \hat{\xi}_4(r_{30}\hat{\xi}_1^3 + r_{21}\hat{\xi}_1^2 + r_{12} + r_{03}), \\
\hat{\zeta}_{21}(\theta) &= \hat{\xi}_3(3k_{30}\hat{\xi}_1^2 + 2k_{21}\hat{\xi}_1\hat{\xi}_2) + \hat{\xi}_4(3r_{30}\hat{\xi}_1^2\hat{\xi}_2 + 2r_{21}\hat{\xi}_1\hat{\xi}_2), \\
\hat{\zeta}_{12}(\theta) &= \hat{\xi}_3(3k_{30}\hat{\xi}_1\hat{\xi}_2^2 + k_{21}\hat{\xi}_2^2) + \hat{\xi}_4(3r_{30}\hat{\xi}_1\hat{\xi}_2^2 + r_{21}\hat{\xi}_2^2), \\
\hat{\zeta}_{03}(\theta) &= \hat{\xi}_2^3(k_{30}\hat{\xi}_3 + r_{30}\hat{\xi}_4).
\end{aligned}$$

(2) The coefficients θ_1 , θ_2 , $\tilde{\sigma}_{ij}(\theta)$, and $\tilde{\zeta}_{ij}(\theta)$ in (3.18):

$$\begin{aligned}
\theta_1 &= \tilde{D}_{21}(1 + \tilde{D}_{12}) - \tilde{D}_{11}\tilde{D}_{22}, \theta_2 = \tilde{D}_{11} + \tilde{D}_{22} - 1, \\
\tilde{\sigma}_{20} &= -\frac{\tilde{D}_{11}}{1 + \tilde{D}_{12}}\hat{\sigma}_{11} + \hat{\sigma}_{20} + \frac{\tilde{D}_{11}^2}{(1 + \tilde{D}_{12})}\hat{\sigma}_{02}, \quad \tilde{\sigma}_{11} = \frac{1}{1 + \tilde{D}_{12}}\hat{\sigma}_{11} - \frac{2\tilde{D}_{11}}{(1 + \tilde{D}_{12})^2}\hat{\sigma}_{02}, \\
\tilde{\sigma}_{02} &= \frac{1}{(1 + \tilde{D}_{12})^2}\hat{\sigma}_{02}, \quad \tilde{\sigma}_{30} = \hat{\sigma}_{30} - \frac{\tilde{D}_{11}}{1 + \tilde{D}_{12}}\hat{\sigma}_{21} + \frac{\tilde{D}_{12}^2}{(1 + \tilde{D}_{12})^2}\hat{\sigma}_{12} - \frac{\tilde{D}_{11}^3}{(1 + \tilde{D}_{12})^3}\hat{\sigma}_{03}, \\
\tilde{\sigma}_{03} &= \frac{1}{(1 + \tilde{D}_{12})^3}\hat{\sigma}_{03}, \quad \tilde{\sigma}_{21} = \frac{1}{1 + \tilde{D}_{12}}\hat{\sigma}_{21} - \frac{2\theta_1}{(1 + \tilde{D}_{12})^2}\hat{\sigma}_{12} + \frac{3\tilde{D}_{11}^2}{(1 + \tilde{D}_{12})^3}\hat{\sigma}_{03}, \\
\tilde{\sigma}_{12} &= \frac{1}{(1 + \tilde{D}_{12})^2}\hat{\sigma}_{12} - \frac{3\tilde{D}_{11}}{(1 + \tilde{D}_{12})^3}\hat{\sigma}_{03}, \\
\tilde{\zeta}_{20} &= -\frac{\tilde{D}_{11}}{1 + \tilde{D}_{12}}\left[\tilde{D}_{11}\hat{\sigma}_{11} + (1 + \tilde{D}_{12})\hat{\zeta}_{11}\right] + \left[\tilde{D}_{11}\hat{\sigma}_{20} + (1 + \tilde{D}_{12})\hat{\zeta}_{20}\right] \\
&\quad + \frac{\tilde{D}_{11}^2}{(1 + \tilde{D}_{12})^2}\left(\tilde{D}_{11}\hat{\sigma}_{02} + (1 + \tilde{D}_{12})\hat{\zeta}_{02}\right), \\
\tilde{\zeta}_{11} &= \frac{\tilde{D}_{11}}{1 + \tilde{D}_{12}}\left[\tilde{D}_{11}\hat{\sigma}_{11} + (1 + \tilde{D}_{12})\hat{\zeta}_{11}\right] - \frac{2\tilde{D}_{11}}{(1 + \tilde{D}_{12})^2}\left(\tilde{D}_{11}\hat{\sigma}_{02} + (1 + \tilde{D}_{12})\hat{\zeta}_{02}\right), \\
\tilde{\zeta}_{02} &= \frac{1}{(1 + \tilde{D}_{12})^2}\left(\tilde{D}_{11}\hat{\sigma}_{02} + (1 + \tilde{D}_{12})\hat{\zeta}_{02}\right), \quad \tilde{\zeta}_{03} = \frac{1}{(1 + \tilde{D}_{12})^3}\left[\tilde{D}_{11}\hat{\sigma}_{30} + (1 + \tilde{D}_{12})\hat{\zeta}_{30}\right], \\
\tilde{\zeta}_{30} &= \left[\tilde{D}_{11}\hat{\sigma}_{30} + (1 + \tilde{D}_{12})\hat{\zeta}_{30}\right] - \frac{\tilde{D}_{11}}{1 + \tilde{D}_{12}}\left[\tilde{D}_{11}\hat{\sigma}_{21} + (1 + \tilde{D}_{12})\hat{\zeta}_{21}\right] \\
&\quad + \frac{\theta_1^2}{(1 + \tilde{D}_{12})^2}\left[\tilde{D}_{11}\hat{\sigma}_{12} + (1 + \tilde{D}_{12})\hat{\zeta}_{12}\right] - \frac{\tilde{D}_{11}^3}{(1 + \tilde{D}_{12})^3}\left[\tilde{D}_{11}\hat{\sigma}_{03} + (1 + \tilde{D}_{12})\hat{\zeta}_{03}\right], \\
\tilde{\zeta}_{21} &= \left[\tilde{D}_{11}\hat{\sigma}_{21} + (1 + \tilde{D}_{12})\hat{\zeta}_{21}\right] - \frac{2\tilde{D}_{11}}{(1 + \tilde{D}_{12})^2}\left[\tilde{D}_{11}\hat{\sigma}_{12} + (1 + \tilde{D}_{12})\hat{\zeta}_{12}\right] \\
&\quad + \frac{3\tilde{D}_{11}^2}{(1 + \tilde{D}_{12})^3}\left[\tilde{D}_{11}\hat{\sigma}_{30} + (1 + \tilde{D}_{12})\hat{\zeta}_{30}\right], \\
\tilde{\zeta}_{12} &= \frac{1}{(1 + \tilde{D}_{12})^2}\left[\tilde{D}_{11}\hat{\sigma}_{12} + (1 + \tilde{D}_{12})\hat{\zeta}_{12}\right] - \frac{3\tilde{D}_{11}^2}{(1 + \tilde{D}_{12})^3}\left[\tilde{D}_{11}\hat{\sigma}_{30} + (1 + \tilde{D}_{12})\hat{\zeta}_{30}\right].
\end{aligned}$$